

FBISE BIOLOGY - CLASS 9

1st Annual Examination 2026

Complete Solutions of Three Papers

Sections A, B and C • Every OR Part Included

MCQs include the reason for the correct choice and an explanation of why each alternative is incorrect. Subjective answers are written according to the marks shown on the paper.

Prepared by Fakhar STEM Sphere

For learning and examination preparation

PAPER 1 - VERSION 10061 (B)

SECTION A - MCQs (12 marks)

1. Climate-change research involving biology, meteorology and geology mainly aims to:

Correct option: C - Integrate different expertise. The problem crosses disciplinary boundaries, so specialists combine knowledge and methods.

Why the other options are wrong: A contradicts the purpose of research; B would prevent evidence collection; D obstructs cooperation rather than advancing it.

2. Forelimbs of human, whale and bat are evidence of evolution because they are:

Correct option: D - Homologous organs. They share the same basic skeletal plan and evolutionary origin although they perform different functions.

Why the other options are wrong: Analogous organs have similar function but different origin; vestigial organs have lost most ancestral function; "modified organs" is too general and is not the required evidence category.

3. If guanine occurs on one DNA strand, the complementary base is:

Correct option: D - Cytosine. DNA base-pairing is G-C through three hydrogen bonds.

Why the other options are wrong: Adenine pairs with thymine, thymine pairs with adenine, and uracil occurs mainly in RNA rather than DNA.

4. Essential condition for organisms to belong to the same species:

Correct option: C - Produce fertile offspring. Members of a species can naturally interbreed and produce fertile offspring.

Why the other options are wrong: Artificial interbreeding is not the criterion; sharing a habitat is insufficient; identical offspring are not required because variation occurs.

5. DNA replication occurs during which cell-cycle phase?

Correct option: D - S phase. S means synthesis; the DNA is duplicated before division.

Why the other options are wrong: G1 is growth, G2 is preparation after replication, and M is nuclear/cell division.

6. Smooth movement and mixing of food in the stomach is due to:

Correct option: A - Muscular tissue. Smooth muscle contracts to churn food and perform peristalsis.

Why the other options are wrong: Connective tissue supports, nervous tissue coordinates, and epithelial tissue lines/secretates; none produces the main mixing force.

7. Enzymes are mainly made of:

Correct option: B - Proteins. Most enzymes are globular proteins whose active sites bind substrates.

Why the other options are wrong: Lipids and carbohydrates are not ordinary catalysts; nucleic acids include rare catalytic RNA, but this is not true of enzymes generally.

8. Plants obtain mineral nutrients from soil mainly through:

Correct option: A - Root hairs. Root hairs provide a very large surface area for absorption of water and ions.

Why the other options are wrong: Xylem transports absorbed material upward; parenchyma mainly stores/photosynthesizes; phloem transports organic food.

9. Transpiration decreases in high humidity because:

Correct option: C - Air is saturated. Moist air reduces the water-vapour concentration gradient between leaf and atmosphere.

Why the other options are wrong: Air-pressure rise is not the main cause; stomata need not close solely due to humidity; leaf area does not suddenly decrease.

10. Which process involves cloning?

Correct option: B - Tissue culture. Cells from one parent grow by mitosis into genetically identical plantlets.

Why the other options are wrong: Grafting joins two plants; parthenogenesis develops from an unfertilized egg; budding is asexual but "cloning" here specifically refers to tissue-culture multiplication.

11. After double fertilization, the triploid nucleus develops into:

Correct option: A - Endosperm. The second male gamete fuses with two polar nuclei to form the triploid primary endosperm nucleus.

Why the other options are wrong: The ovule becomes the seed; the zygote becomes embryo; integuments form seed coat.

12. Cell responsible for oxygen transport in animals:

Correct option: B - Red blood cell. Its haemoglobin binds and carries oxygen.

Why the other options are wrong: Muscle cells contract, nerve cells conduct impulses, and guard cells regulate stomata in plants.

SECTION B - Short Questions (all OR parts solved)

2(i). How are RNA types involved in converting hereditary information into functional proteins?

mRNA carries the copied genetic code (codons) from DNA to a ribosome. rRNA forms the main structural/catalytic part of the ribosome and helps make peptide bonds. tRNA brings the correct amino acid and pairs its anticodon with the mRNA codon. Thus DNA information is translated into an amino-acid sequence.

2(i) OR. Saturated vs unsaturated fatty acids; basic lipid structure

Saturated: no C=C bond; usually straight and solid at room temperature. Unsaturated: one or more C=C bonds; kinked and commonly liquid. A typical triglyceride consists of one glycerol joined to three fatty acids by ester bonds.

2(ii). Briefly explain speciation.

Speciation is the formation of a new species from an existing population. Geographical/reproductive isolation stops gene flow; mutation, natural selection and genetic drift accumulate differences. Eventually the populations can no longer interbreed to produce fertile offspring.

2(ii) OR. Biotechnology and biophysics with examples

Biotechnology uses organisms/cells or their products to make useful products; e.g., recombinant bacteria producing human insulin. Biophysics applies physical laws and instruments to biological systems; e.g., studying blood flow or using X-ray diffraction to investigate biomolecules.

2(iii). Why are starch and cellulose polysaccharides? Give two characteristics.

Both are long polymers made by joining many glucose monosaccharides through glycosidic bonds. Polysaccharides are generally large/complex and insoluble or only slightly soluble in water; they serve storage (starch) or structural (cellulose) roles.

2(iii) OR. How do runners help vegetative propagation?

A runner is a horizontal stem that grows over the soil. At nodes it forms adventitious roots and buds/shoots. Each rooted node develops into an independent, genetically identical plant; strawberry and grass are examples.

2(iv). Effect of temperature on transpiration

Increasing temperature gives water molecules more kinetic energy, raises evaporation from mesophyll surfaces and increases the diffusion gradient, so transpiration usually rises. Very high temperature or water shortage may close stomata and reduce the rate.

2(iv) OR. Three plant macronutrients and roles

Nitrogen: amino acids, proteins and chlorophyll. Phosphorus: ATP, nucleic acids and membranes; supports roots. Potassium: enzyme activation, osmotic balance and stomatal movement.

2(v). Domain containing organisms without a true nucleus

Bacteria (also Archaea are prokaryotic): cells lack a membrane-bound nucleus and membrane-bound organelles; DNA is circular in a nucleoid; organisms are unicellular and divide by binary fission. Bacterial cell walls usually contain peptidoglycan.

2(v) OR. Roles of epithelial and connective tissue in stomach

Epithelial tissue lines the stomach: mucus protects it and gastric glands secrete acid/enzymes. Connective tissue supports and binds the layers and contains blood vessels, nerves and immune cells.

2(vi). How comparative anatomy supports evolution

Homologous organs (human arm, whale flipper, bat wing) have the same basic plan but different functions, indicating common ancestry and divergent evolution. Analogous organs show similar environmental selection, while vestigial organs are remnants inherited from ancestors.

2(vi) OR. Family, genus and class in taxonomy

Family: related genera (Felidae). Genus: closely related species (Panthera). Class: related orders (Mammalia). Rank order from broader to narrower is Class → Order → Family → Genus → Species.

2(vii). Why many mitochondria benefit muscle cells

Contraction repeatedly requires ATP for actin-myosin cycling, ion pumps and recovery. Numerous mitochondria provide a large surface for aerobic respiration and generate ATP rapidly, especially in active muscle.

2(vii) OR. Why grafting is artificial propagation and improves fruit quickly

Humans deliberately attach a desired scion to a rooted stock; therefore it is artificial vegetative propagation. The mature superior scion retains its fruit traits and bears earlier, while the stock can add vigour and disease resistance.

2(viii). Effect of a non-competitive inhibitor

It binds at an allosteric site rather than the active site and changes enzyme shape/activity. Fewer functional enzyme molecules remain, so maximum rate decreases; adding more substrate cannot fully overcome the inhibition.

2(viii) OR. Mitosis in asexual reproduction and cell replacement

Mitosis gives genetically identical daughter cells and preserves chromosome number. It produces clones in asexual reproduction and replaces worn-out/dead cells for growth, repair and healing.

2(ix). Histology, genetics and pharmacology

Histology: microscopic study of tissues. Genetics: study of heredity and variation. Pharmacology: study of drugs, their actions, uses and effects on organisms.

2(ix) OR. Three differences: meiosis vs mitosis

Meiosis has two divisions while mitosis has one; meiosis produces four haploid, genetically varied cells while mitosis produces two diploid, nearly identical cells; homologous pairing/crossing-over occurs in prophase I but not in normal mitosis.

2(x). Identify labelled leaf cross-section

P: stoma/guard cells - controls gas exchange and transpiration. Q: palisade mesophyll - tightly packed, many chloroplasts, main photosynthesis. R: spongy mesophyll - loose cells/air spaces, gas diffusion (and some photosynthesis). S: upper epidermis. T: vascular bundle (xylem/phloem).

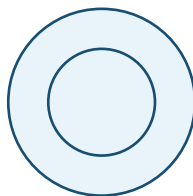
2(x) OR. Heart: level of organization and four examples

The heart is an organ because several tissues cooperate for one major function. Other organ-level examples: stomach, kidney, lung and leaf.

SECTION C - Long Questions (all OR parts solved)

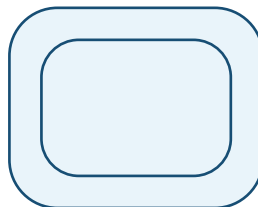
Q3. Sketch and label lysosome, vacuole and centriole; one role each

- Lysosome: small single-membrane vesicle containing hydrolytic enzymes; digests worn-out organelles, microbes and macromolecules.
- Vacuole: membrane (tonoplast) surrounding cell sap; stores water/solutes and maintains turgor in plants.
- Centriole: cylinder of nine microtubule triplets; organizes spindle microtubules during animal-cell division.
- In the exam, draw three separate, neat labelled figures and write the function below each.



LYSOSOME

single membrane + enzymes



VACUOLE

tonoplast / cell sap



CENTRIOLES

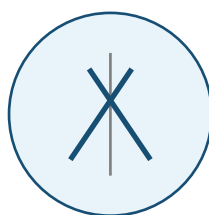
9 microtubule triplets

Q3 OR. How Meiosis I reduces chromosome number; stages and diagrams

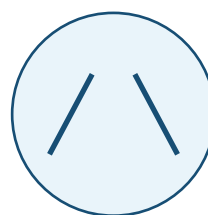
- Before meiosis, DNA replicates once. In prophase I, homologous chromosomes pair (synapsis) as bivalents; crossing-over may occur; nuclear envelope disappears and spindle forms.
- In metaphase I, bivalents align independently at the equator, with homologues attached to opposite poles.
- In anaphase I, homologous chromosomes separate; centromeres do not divide, so sister chromatids remain together.
- In telophase I/cytokinesis, two cells/nuclei form. Each receives one chromosome of every homologous pair: $2n$ becomes n , although every chromosome still has two chromatids.
- Reduction therefore occurs because homologues, not sister chromatids, are separated in the first division.



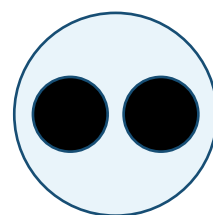
Prophase I



Metaphase I



Anaphase I



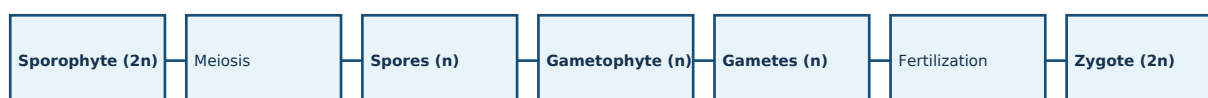
Telophase I

Q4. Types and significance of anaerobic respiration

- Anaerobic respiration releases energy from glucose without oxygen and occurs in cytoplasm. It yields only 2 ATP per glucose.
- Alcoholic fermentation (yeast/plants): $\text{glucose} \rightarrow 2 \text{ ethanol} + 2 \text{ CO}_2 + 2 \text{ ATP}$.
- Lactic-acid fermentation (muscles/some bacteria): $\text{glucose} \rightarrow 2 \text{ lactic acid} + 2 \text{ ATP}$.
- Significance: supplies rapid ATP when oxygen is scarce; yeast is used in bread (CO_2 raises dough) and brewing; bacteria make yoghurt/cheese; exercising muscles can continue briefly, after which oxygen helps remove/oxidize lactate.

Q4 OR. Male gametophyte, seed parts and plant life cycle

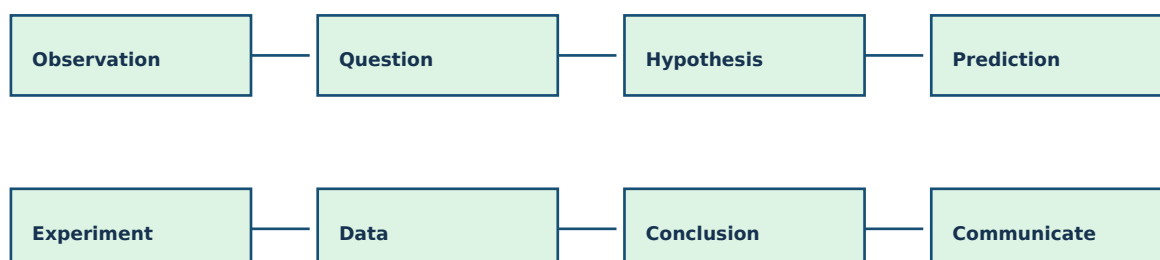
- Inside anther, diploid microspore mother cells undergo meiosis to form haploid microspores. Each microspore undergoes mitosis and becomes a pollen grain with tube cell and generative cell; the generative cell divides to form two male gametes.
- After pollination the pollen tube grows through the style; tube nucleus guides it and the two male gametes enter the embryo sac.
- A seed consists of seed coat (protection), embryo (radicle, plumule and cotyledon/s) and food reserve (endosperm/cotyledons).
- Life cycle: diploid sporophyte → meiosis → haploid spores → male/female gametophytes → gametes → fertilization → diploid zygote → embryo/seed → germination → sporophyte.



Alternation of generations: haploid and diploid phases

Q5. Scientific method for solving a biological problem

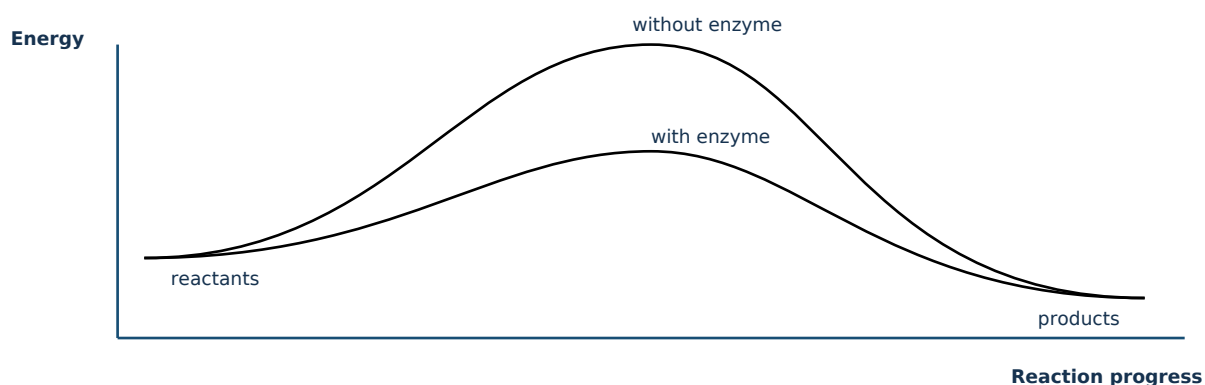
- Observation: notice and record a phenomenon, e.g., malaria is common near stagnant water.
- Problem/question: ask a focused, testable question: does mosquito exposure transmit malaria?
- Hypothesis: propose a testable explanation: a particular mosquito carries the parasite from patient to healthy person.
- Deduction/prediction: if true, protected people should have fewer infections and the parasite should occur in the vector.
- Experiment: compare experimental and control groups; change one independent variable, measure the dependent variable and keep controlled variables constant; use adequate sample/replication and ethical precautions.
- Results/data: record in tables/graphs and analyse statistically.
- Conclusion: accept, reject or revise the hypothesis according to evidence.
- Communication/repetition: publish methods/results for peer review and independent verification; repeated support can contribute to a theory.



Revise hypothesis and repeat when evidence disagrees

Q5 OR. Types of enzyme cofactors and effect on activation energy

- A cofactor is a non-protein helper required by some enzymes. Apoenzyme + cofactor forms the active holoenzyme.
- Inorganic activators: metal ions such as Mg^{2+} , Zn^{2+} or Fe^{2+} that stabilize enzyme/substrate or participate in reaction.
- Organic coenzymes: small, loosely bound carriers, often vitamin-derived, e.g., NAD^+ and coenzyme A.
- Prosthetic groups: organic groups tightly/permanently attached, e.g., haem or FAD in some enzymes.
- Enzymes provide an alternative pathway with lower activation energy. They increase rate but do not change reactant/product energies, overall ΔG , or equilibrium. On an energy graph, the catalysed peak is lower than the uncatalysed peak.



PAPER 2 - VERSION 12061 (D)

SECTION A - MCQs (12 marks)

1. Best evidence of scientific collaboration in a mixed research team:

Correct option: C - Shared analysis and feedback. Sharing results and critically reviewing one another integrates expertise and improves reliability.

Why the other options are wrong: Working alone, secrecy and accepting one opinion prevent collaboration and independent checking.

2. Fossils of the same organism in different rock layers mainly show:

Correct option: C - Gradual change over time. Successive strata preserve changes in organisms across geological time.

Why the other options are wrong: They do not support sudden creation, absence of variation, or merely random blockage.

3. Complementary DNA strand for A-T-G-C-T:

Correct option: B - T-A-C-G-A. DNA pairing is A-T and G-C, so every base is replaced by its complement.

Why the other options are wrong: A repeats the original; C contains uracil and is not DNA; D does not follow complementary pairing.

4. A species is best defined as organisms that:

Correct option: B - Interbreed naturally. They reproduce naturally with one another and produce fertile offspring.

Why the other options are wrong: Appearance, living together or sharing habitat alone cannot establish the biological species boundary.

5. In G1 phase of the cell cycle:

Correct option: A - Nucleotides are formed. The cell grows and synthesizes materials needed for later DNA replication.

Why the other options are wrong: DNA replicates in S; exit to G0 may occur but is not the defining event; division occurs in M.

6. Main stomach tissue responsible for secretion of gastric juice:

Correct option: A - Epithelial. Glandular epithelial cells in gastric glands secrete acid, enzymes and mucus.

Why the other options are wrong: Nervous tissue coordinates, muscle churns, and connective tissue supports.

7. An enzyme is best described as a/an:

Correct option: D - Biological catalyst. It increases reaction rate without being consumed by lowering activation energy.

Why the other options are wrong: It is usually a protein, not a lipid, hormone or energy source.

8. Magnesium is required for synthesis of:

Correct option: A - Chlorophyll. Mg is the central ion of the chlorophyll molecule.

Why the other options are wrong: Cell wall is mainly cellulose; guard cells are cells, not a Mg product; proteins mainly require amino-acid elements such as nitrogen.

9. Structure NOT related to xylem tissue:

Correct option: B - Sieve tubes. Sieve-tube elements belong to phloem.

Why the other options are wrong: Vessels, tracheids and xylem fibres are xylem components.

10. Cloning produces:

Correct option: C - Identical cells. Clones arise mitotically from one parent and are genetically identical except for mutations.

Why the other options are wrong: Different or mutated cells are not the intended products, and hybrid species require genetic mixing.

11. After pollination, the process required for seed formation is:

Correct option: D - Fertilization. Fusion of gametes produces the zygote; ovule development then forms the seed.

Why the other options are wrong: Germination and dispersal occur after seed formation; meiosis produced spores/gametes earlier.

12. Leaf cells containing the most chloroplasts:

Correct option: D - Mesophyll cells. Especially palisade mesophyll, they are the main photosynthetic cells.

Why the other options are wrong: Epidermis usually lacks chloroplasts except guard cells; sclerenchyma is mainly dead support tissue; generic parenchyma may store rather than photosynthesize.

SECTION B - Short Questions (all OR parts solved)

2(i). Identify biomolecules P and Q and one function each

P is RNA: usually single-stranded; it carries/decodes genetic information during protein synthesis. Q is DNA: double helix; it stores and transmits hereditary information.

2(i) OR. Complete taxonomic ranks

Domain → Kingdom (II) → Phylum (III) → Class (IV) → Order (V) → Family (VI) → Genus (VII) → Species (VIII).

2(ii). Main points of Darwin's theory and key principle

Populations overproduce; individuals show heritable variation; limited resources cause struggle; better-adapted individuals survive/reproduce more; favourable traits accumulate. The key principle is natural selection ("differential reproductive success"), producing adaptation over generations.

2(ii) OR. Three biology careers

Microbiologist: studies microorganisms. Geneticist: studies genes, heredity and variation. Ecologist: studies interactions among organisms and environment.

2(iii). Monosaccharides vs disaccharides

Monosaccharides contain one sugar unit; disaccharides contain two. Monosaccharides cannot be hydrolysed into simpler carbohydrates; disaccharides hydrolyse into two monosaccharides. Examples: glucose/fructose versus sucrose/maltose.

2(iii) OR. Parthenogenesis and why it is sexual

It is development of an individual from an unfertilized egg (e.g., male honeybee). It is classed as a special sexual mode because an egg/gamete is produced, although fertilization and fusion with a male gamete do not occur.

2(iv). Wind and humidity effects on transpiration

Wind removes the humid boundary layer, increasing the water-vapour gradient and transpiration (extreme wind may cause stomatal closure). High humidity lowers the gradient and reduces transpiration; low humidity increases it.

2(iv) OR. Macro- and micronutrients

Macronutrients are required in relatively large amounts, e.g., nitrogen for proteins/chlorophyll. Micronutrients are needed in traces, e.g., iron for electron-transfer enzymes and chlorophyll formation.

2(v). Three domains and a feature each

Bacteria: prokaryotic, peptidoglycan wall. Archaea: prokaryotic, wall lacks peptidoglycan and many live in extreme habitats. Eukarya: cells possess a true nucleus and membrane-bound organelles.

2(v) OR. Three stomach tissues and functions

Epithelial tissue lines/protects and secretes gastric juice; muscular tissue churns food and performs peristalsis; connective tissue binds/supports layers and carries blood vessels/nerves.

2(vi). How paleontology supports evolution

Fossils in dated rock layers provide a chronological record. Older layers contain ancestral forms; younger layers show modified descendants and transitional fossils, supporting descent with change and extinction.

2(vi) OR. Emergent properties from organelles to tissues

An emergent property appears through interaction of parts and is absent from a part alone. Organelles cooperate to make a functioning cell; similar specialized cells plus extracellular material cooperate as tissue. For example, many muscle cells coordinated together generate organized contraction greater than any organelle can produce.

2(vii). Why a plant cell does not burst in hypotonic solution

Water enters by osmosis and the vacuole enlarges, but the rigid cellulose cell wall resists expansion. Wall pressure balances turgor pressure, making the cell turgid rather than lysed.

2(vii) OR. Stem cuttings: type and reasons

This is artificial vegetative propagation. Cuttings rapidly make clones with desired traits, avoid dependence on seeds/pollination, allow propagation of seedless plants, and often flower/fruit earlier.

2(viii). Competitive inhibition

A substrate-like inhibitor reversibly occupies the active site and competes with substrate. It reduces reaction rate at a given substrate concentration, but its effect can be reduced by adding excess substrate.

2(viii) OR. Meiosis and maintenance of chromosome number

Meiosis halves the diploid number to haploid in gametes/spores. Fertilization restores diploid number, so each generation retains the species chromosome number in animals and in the sexual phase of plants.

2(ix). Biometry and biogeography

Biometry: statistical/mathematical analysis of biological data, e.g., mean plant height. Biogeography: study of organism distribution through space/time, e.g., island species distribution.

2(ix) OR. Stem cells and types

Stem cells are unspecialized cells able to self-renew and differentiate. Embryonic stem cells are pluripotent and form nearly all body cell types; adult/somatic stem cells are usually multipotent and repair a limited tissue range.

2(x). Three main tissues in a typical leaf

Epidermal tissue protects and limits water loss; mesophyll tissue carries out photosynthesis/gas exchange; vascular tissue (xylem/phloem) transports water/minerals and food.

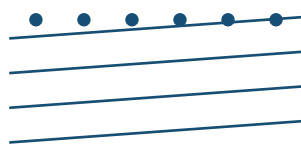
2(x) OR. Lipids and basic structure

Lipids are hydrophobic organic biomolecules, insoluble in water but soluble in organic solvents. A common neutral lipid is a triglyceride: one glycerol molecule ester-bonded to three fatty acids.

SECTION C - Long Questions (all OR parts solved)

Q3. Draw ER, Golgi apparatus and mitochondrion; one role each

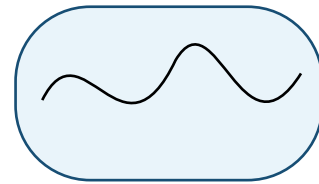
- Endoplasmic reticulum: interconnected membranes; rough ER bears ribosomes and makes/transport proteins, while smooth ER makes lipids and detoxifies.
- Golgi apparatus: stacked flattened cisternae with vesicles; modifies, sorts and packages proteins/lipids for secretion or other destinations.
- Mitochondrion: double membrane with inner cristae and matrix; performs aerobic respiration and produces most cellular ATP.
- Label membrane/cisternae, ribosomes where appropriate, vesicles, cristae and matrix.



ROUGH ER
cisternae + ribosomes



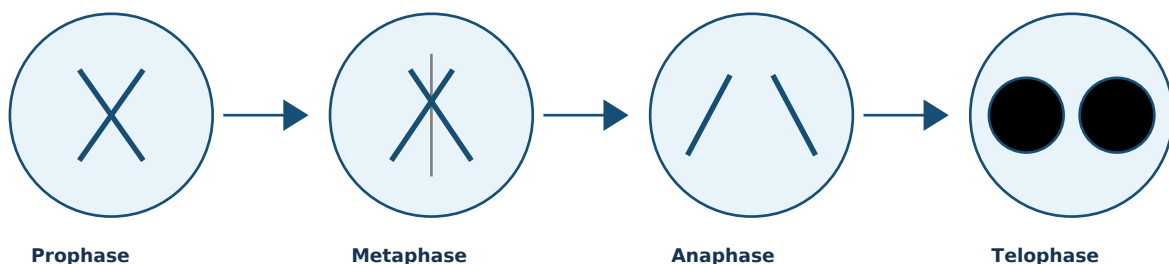
GOLGI
cisternae + vesicles



MITOCHONDRION
cristae / matrix

Q3 OR. Karyokinesis in mitosis with labelled diagram

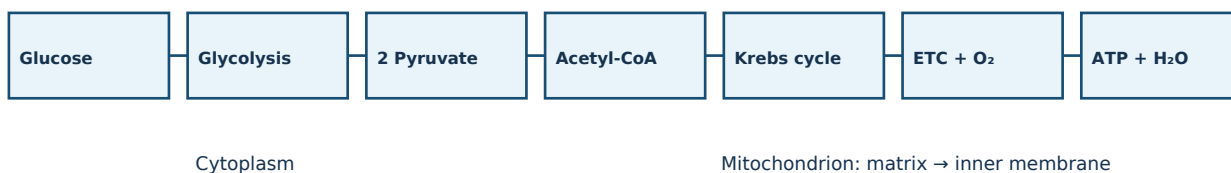
- Prophase: replicated chromatin condenses into chromosomes (two sister chromatids); centrosomes move to poles, spindle forms, nucleolus and nuclear envelope disappear.
- Metaphase: chromosomes align at the equator and spindle fibres attach to kinetochores from opposite poles.
- Anaphase: centromeres divide; sister chromatids separate and move to opposite poles as daughter chromosomes.
- Telophase: chromosomes reach poles and decondense; nuclear envelopes/nucleoli re-form and spindle disappears. Cytokinesis then divides cytoplasm.
- The two daughter nuclei are genetically alike and retain the parental chromosome number.



Q4. Steps of aerobic respiration and chemical equation

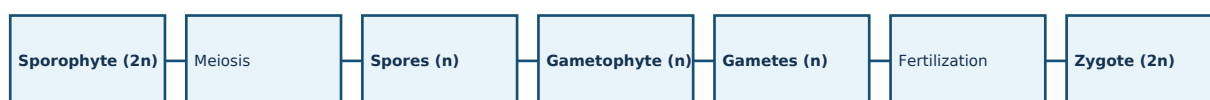
- Overall: $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{energy (ATP + heat)}$.
- Glycolysis (cytoplasm): glucose $\rightarrow$ 2 pyruvate; net 2 ATP and 2 NADH; oxygen is not used directly.
- Link reaction (mitochondrial matrix): each pyruvate $\rightarrow$ acetyl-CoA + CO_2 ; NADH forms.
- Krebs cycle (matrix): acetyl-CoA is fully oxidized; CO_2 , a little ATP, NADH and FADH_2 form.

- Electron-transport chain/oxidative phosphorylation (inner membrane): electrons from NADH/FADH₂ drive proton pumping; H⁺ flows through ATP synthase. Oxygen is the final electron acceptor and forms water; most ATP is produced here.



Q4 OR. Female gametophyte, double fertilization and plant life cycle

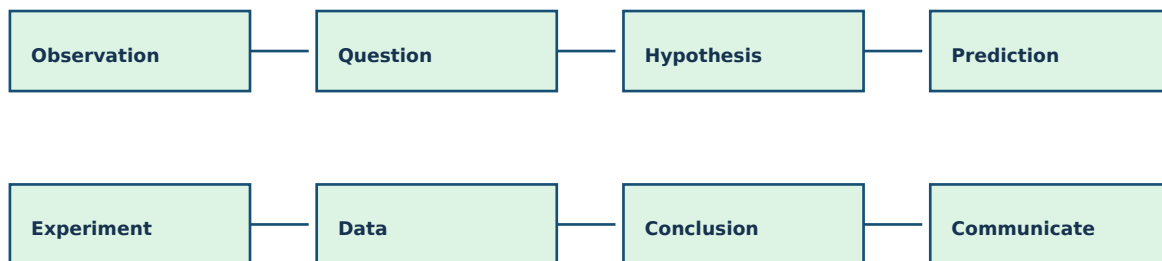
- In the ovule, a diploid megaspore mother cell undergoes meiosis; usually one haploid megaspore survives. Three mitoses form an eight-nucleate embryo sac: egg + two synergids, three antipodals and two polar nuclei in the central cell.
- A pollen tube enters a synergid and releases two male gametes. One fuses with egg → diploid zygote (syngamy); the other fuses with two polar nuclei → triploid primary endosperm nucleus (triple fusion). Together these are double fertilization.
- The zygote forms embryo; endosperm nourishes it; integuments form seed coat; ovule becomes seed and ovary becomes fruit.
- Alternation: sporophyte (2n) → meiosis/spores (n) → gametophytes (n) → gametes → fertilization/zygote (2n) → new sporophyte.



Alternation of generations: haploid and diploid phases

Q5. Scientific method for solving a biological problem

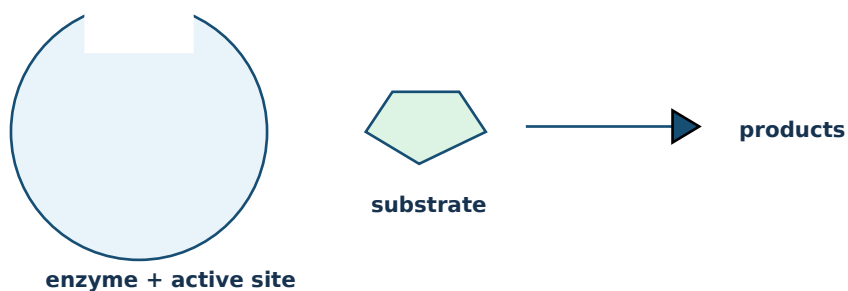
- Observation and background research identify a pattern; a precise, testable question is stated.
- A hypothesis offers a tentative explanation and deductions predict results if it is correct.
- A controlled experiment compares control and experimental groups, changes only the independent variable, measures a dependent variable, controls other factors, and uses replication/sample size.
- Data are recorded objectively, displayed in tables/graphs and analysed. Results lead to a conclusion that supports, rejects or modifies the hypothesis.
- Scientists communicate procedures/results for peer review and repetition. Consistent evidence builds reliable explanations and may support a theory.



Revise hypothesis and repeat when evidence disagrees

Q5 OR. What are enzymes? Explain five characteristics

- Enzymes are biological catalysts, mostly globular proteins, that speed reactions by lowering activation energy without being consumed.
- Specificity: an active site binds particular substrate(s).
- Reusable/effective in small amount: unchanged at reaction end.
- Optimum conditions: each has an optimum temperature and pH; extremes reduce activity/denature protein.
- Rate dependence: rate increases with substrate concentration until active sites are saturated.
- Regulation: inhibitors reduce and activators/cofactors assist activity.
- They do not alter equilibrium or ΔG ; many catalyse reversible reactions and are commonly named with “-ase”.



PAPER 3 - VERSION 19061 (M)

SECTION A - MCQs (12 marks)

1. Yellowing of leaves commonly occurs due to deficiency of:

Correct option: A - Magnesium. Mg is part of chlorophyll; deficiency causes chlorosis, often first in older leaves.

Why the other options are wrong: Calcium deficiency affects growing points, iron chlorosis usually begins in young leaves, and manganese has distinct interveinal/mottled symptoms; the curriculum answer here is Mg.

2. Strong wind increases transpiration mainly by:

Correct option: C - Moving water vapours. Wind removes the humid boundary layer, maintaining a steep diffusion gradient.

Why the other options are wrong: Cooling can lower transpiration; reduced light tends to close stomata; wind does not increase rate mainly by directly opening stomata.

3. Tissue culture is an example of:

Correct option: D - Cloning. Mitotic growth from an explant produces genetically identical plantlets.

Why the other options are wrong: It is not sexual reproduction, budding or grafting.

4. In flowering plants, the zygote develops into:

Correct option: A - Embryo. The diploid zygote divides mitotically to form the embryo.

Why the other options are wrong: The ovule becomes seed, the primary endosperm nucleus forms endosperm, and seed includes embryo plus coverings/food reserve.

5. Cell specialized for transmission of nerve impulses:

Correct option: B - Neuron. Its dendrites, cell body and long axon receive and conduct impulses.

Why the other options are wrong: Guard cells regulate stomata, root hairs absorb, and Amoeba is a whole unicellular organism.

6. Scientists criticizing one another's methods mainly helps to:

Correct option: C - Improve accuracy. Peer criticism exposes weaknesses, improves procedures and makes conclusions more reliable.

Why the other options are wrong: It is not intended to delay, create conflict or limit discussion.

7. Development of dog breeds with different traits is:

Correct option: D - Selective breeding. Humans choose parents with desired characteristics over generations.

Why the other options are wrong: Natural selection is environmental, genetic drift is random frequency change, and mutation alone does not describe planned breeding.

8. mRNA transcribed from DNA template A-T-C-G-T is:

Correct option: A - U-A-G-C-A. RNA pairs A-U, T-A, C-G and G-C with the DNA template.

Why the other options are wrong: B uses thymine instead of uracil; C and D contain incorrect complementary bases.

9. Artificial crossing of two different species usually produces:

Correct option: B - Sterile offspring. Interspecific hybrids often have mismatched chromosome sets and cannot form balanced gametes.

Why the other options are wrong: Fertility is uncommon; offspring are not identical; similarity to one parent does not define the key outcome.

10. Usually the longest phase of the cell cycle:

Correct option: C - Interphase. G1, S and G2 together occupy most of the cycle for growth and replication.

Why the other options are wrong: M phase is shorter; G2 and S are only portions of interphase.

11. Tissue supporting and binding other stomach tissues:

Correct option: D - Connective. Its extracellular matrix binds, supports and carries vessels/nerves.

Why the other options are wrong: Muscle contracts, epithelium lines/secretates, and nervous tissue coordinates.

12. An organic molecule permanently attached to an enzyme is a/an:

Correct option: B - Prosthetic group. A prosthetic group is a tightly or permanently bound organic cofactor.

Why the other options are wrong: An activator is commonly an ion; a coenzyme binds loosely; cofactor is the broader category and may be inorganic.

SECTION B - Short Questions (all OR parts solved)

2(i). Structural advantages of epidermal cell and red blood cell

Plant epidermal cells form a tightly packed transparent layer and often secrete waxy cuticle, protecting while allowing light through. A red blood cell is biconcave, flexible and lacks a nucleus, giving large surface area and more room for haemoglobin for oxygen transport.

2(i) OR. Natural vs artificial vegetative propagation; advantage

Natural propagation uses plant structures without deliberate human action (runner, bulb, rhizome); artificial propagation is performed by humans (cutting, grafting, tissue culture). Natural methods use naturally formed organs, while artificial methods are selected/controlled. Advantage: rapid production of many genetically identical plants with desirable traits.

2(ii). Competitive vs non-competitive inhibition

Competitive inhibitor binds the active site, resembles/competes with substrate and can be overcome by excess substrate. Non-competitive inhibitor binds elsewhere, alters enzyme function, lowers maximum rate and cannot be overcome by adding substrate.

2(ii) OR. Why meiosis produces variation

Crossing-over in prophase I exchanges DNA between homologues; independent assortment randomly aligns maternal/paternal homologues at metaphase I; segregation makes genetically different haploid gametes. Random fertilization later increases combinations.

2(iii). Taxonomy, immunology and morphology with examples

Taxonomy: naming/classifying organisms (classifying humans as *Homo sapiens*). Immunology: immune system and defence (antibody response to vaccine). Morphology: external form/structure (comparing leaf shapes).

2(iii) OR. Prophase of mitosis vs prophase I

Mitosis: no synapsis, no crossing-over, and comparatively short/simple. Prophase I: homologues pair as bivalents, crossing-over/chiasmata occur, and it has five sub-stages and is longer.

2(iv). Three osmotic adjustments in xerophytes

They accumulate compatible solutes (sugars/proline) to lower water potential and retain water; store water/mucilage in succulent tissues; and regulate ions/close stomata through ABA so cells maintain turgor during drought.

2(iv) OR. Two dietary lipid sources and importance

Sources: vegetable oils/nuts and animal fats/eggs/dairy. Lipids provide concentrated energy, form membranes, insulate/protect organs, help make hormones and enable absorption of vitamins A, D, E and K.

2(v). Complete amino-acid structure and name functional groups

The central carbon bonds to H, variable R group, amino group ($-NH_2$), and carboxyl group ($-COOH$). The two functional groups are amino and carboxyl.

2(v) OR. Identify and explain meiotic event

The figure shows crossing-over in prophase I. Homologous chromosomes pair; non-sister chromatids break and rejoin at chiasmata, exchanging equivalent segments and producing recombinant chromatids/variation.

2(vi). Darwin's natural selection

Populations show heritable variation and produce more young than can survive. Competition follows; individuals with favourable traits leave more offspring. Across generations those alleles become common, causing adaptation and possibly new species.

2(vi) OR. Computational biology and biochemistry

Computational biology uses algorithms/computers to analyse biological data, e.g., comparing DNA sequences. Biochemistry studies chemical molecules/reactions in life, e.g., enzyme action in cellular respiration.

2(vii). Sucrose vs maltose

Sucrose = glucose + fructose; maltose = glucose + glucose. Sucrose has an $\alpha 1 \rightarrow \beta 2$ bond and is non-reducing; maltose has an $\alpha 1 \rightarrow 4$ bond and is reducing. Sucrose is common transport/table sugar; maltose forms during

starch digestion/germination.

2(vii) OR. Parthenogenesis with example

It differs from ordinary sexual reproduction because an unfertilized egg develops without fusion of male and female gametes. In honeybees, unfertilized haploid eggs form male drones; fertilized eggs form females.

2(viii). Two advantages and one disadvantage of transpiration

Advantages: creates transpiration pull for water/mineral transport and cools leaves; it also supplies water for photosynthesis/turgor. Disadvantage: excessive loss causes wilting, stomatal closure, reduced photosynthesis or death.

2(viii) OR. Importance of mesophyll tissues

Palisade mesophyll has tightly packed, chloroplast-rich cells for maximum photosynthesis. Spongy mesophyll has loose cells and large air spaces for rapid CO_2/O_2 diffusion and also performs some photosynthesis.

2(ix). Groups in Eukarya and two features

Groups/kingdoms include Protista, Fungi, Plantae and Animalia. Cells have a membrane-bound nucleus with linear chromosomes and membrane-bound organelles such as mitochondria; division uses mitosis/meiosis.

2(ix) OR. Three stomach tissues and functions

Epithelial: lining/protection and secretion. Muscular: churning and peristalsis. Connective: support/binding and passage for vessels/nerves.

2(x). Vestigial organs and selective breeding as evolution evidence

Vestigial organs are reduced ancestral structures (human appendix/coccyx), indicating common descent. Selective breeding shows that selecting heritable variation can greatly change populations over generations, demonstrating evolutionary capacity.

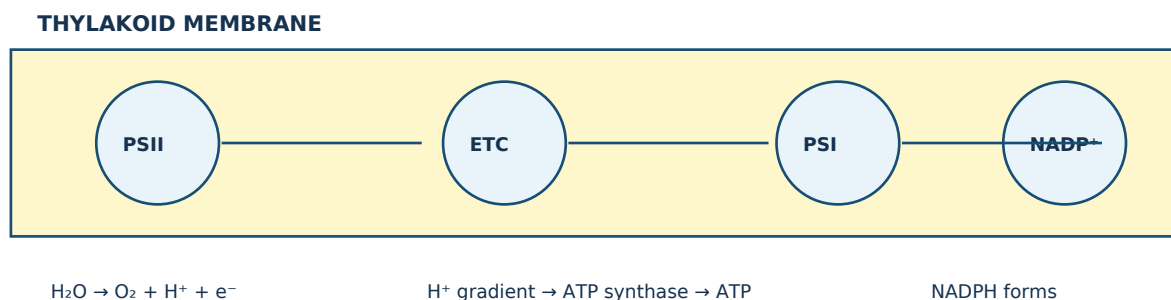
2(x) OR. Highest and lowest modern taxonomic ranks

Domain is highest and contains organisms sharing the broadest cellular/molecular features. Species is lowest: similar organisms that naturally interbreed and produce fertile offspring.

SECTION C - Long Questions (all OR parts solved)

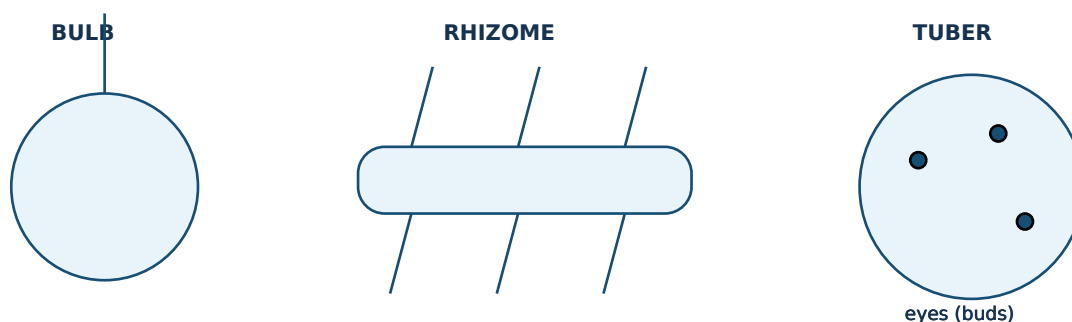
Q3. Light-dependent reactions of photosynthesis

- They occur on thylakoid membranes. Light excites chlorophyll in photosystem II (PSII); high-energy electrons enter an electron-transport chain.
- Water is split (photolysis): $2\text{H}_2\text{O} \rightarrow \text{O}_2 + 4\text{H}^+ + 4\text{e}^-$. Oxygen is released and electrons replace those lost by PSII.
- Electron transfer pumps H^+ into the thylakoid lumen. H^+ returns through ATP synthase, producing ATP by photophosphorylation.
- Light re-excites electrons in photosystem I (PSI); NADP⁺ reductase uses electrons and H^+ to reduce NADP⁺ to NADPH.
- Products are ATP and NADPH for the Calvin cycle, and O_2 as a by-product.



Q3 OR. Vegetative propagation through bulb, rhizome and tuber

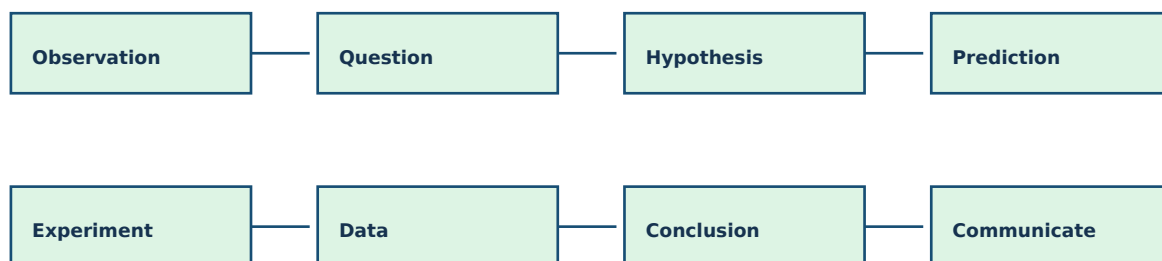
- Bulb: short stem with fleshy food-storing leaves; buds form new shoots/bulbs. Example: onion.
- Rhizome: horizontal underground stem with nodes, buds and adventitious roots; each segment can form a shoot. Example: ginger.
- Stem tuber: swollen underground stem; each “eye” is a bud that grows into a new plant. Example: potato.
- All are natural asexual propagation; offspring are clones of the parent. In diagrams label bud, stem, storage tissue/leaves and roots.



Q4. Scientific method for solving a biological problem

- Observe a biological phenomenon and use background knowledge to define a focused question.
- Form a testable/falsifiable hypothesis and derive measurable predictions.
- Design a fair experiment: control group, experimental group, independent variable, dependent variable, controlled factors, suitable sample and repeated trials.
- Collect quantitative/qualitative data accurately; organize tables and graphs; analyse patterns/statistics.

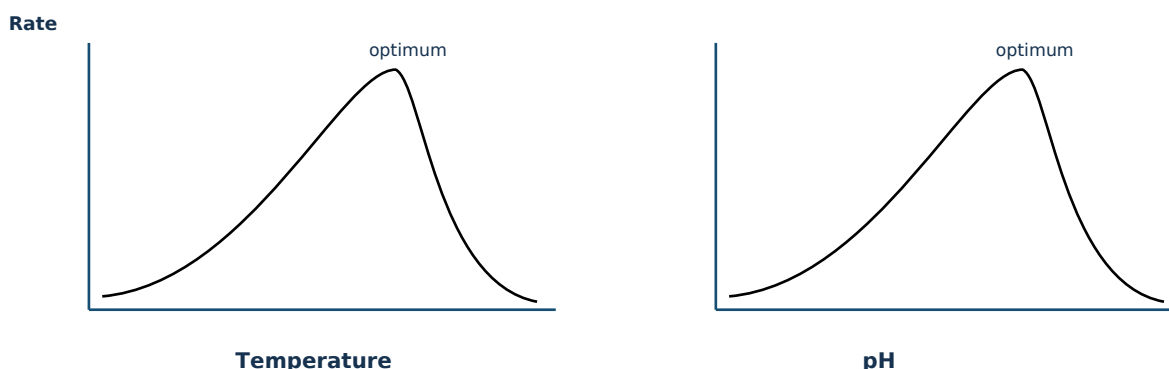
- Draw a conclusion based only on evidence, then accept, reject or revise the hypothesis.
- Report methods and findings for peer review and replication. Repeated evidence increases confidence and contributes to models/theories.



Revise hypothesis and repeat when evidence disagrees

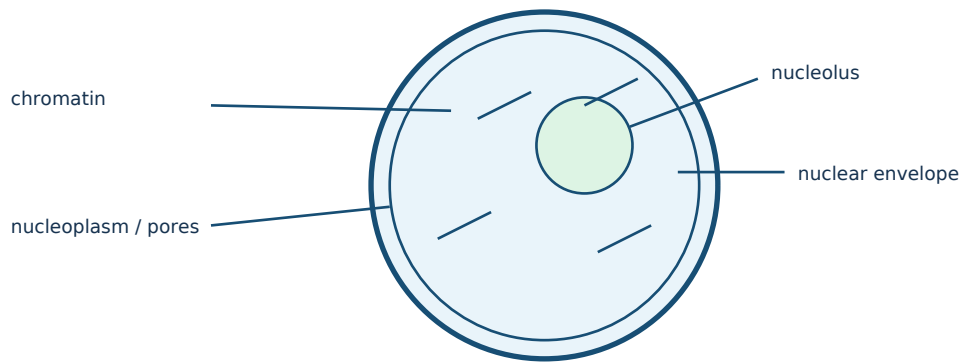
Q4 OR. Effects of temperature and pH on enzyme activity

- At low temperature, molecules move slowly, so few successful enzyme-substrate collisions occur. As temperature rises, rate rises to an optimum. Above it, bonds maintaining enzyme shape break, the active site changes (denaturation) and rate falls sharply.
- pH changes ionization/charges and bonds in the enzyme. Each enzyme has an optimum pH; movement away lowers binding/catalysis and extreme pH denatures it. Examples: pepsin works best near pH 2, while trypsin works best in alkaline small intestine.
- On graphs: temperature gives a rise followed by a steep fall; pH usually gives a bell-shaped curve around the optimum.



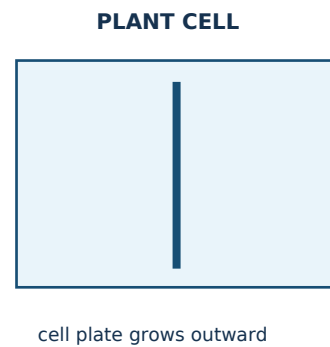
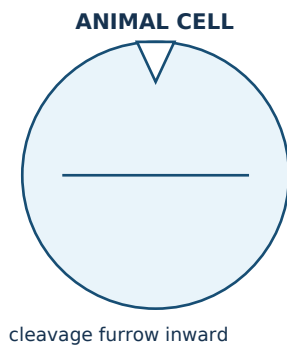
Q5. Draw and label nucleus; functions of its parts

- Nuclear envelope: double membrane separating genetic material from cytoplasm; outer membrane may join rough ER.
- Nuclear pores: selectively exchange RNA, ribosomal subunits and proteins.
- Nucleoplasm: internal fluid supporting chromatin and nuclear reactions.
- Chromatin/chromosomes: DNA plus proteins; stores genes, directs protein synthesis and replicates before division.
- Nucleolus: makes rRNA and assembles ribosomal subunits.
- Overall, the nucleus controls cell activities by regulating gene expression and transmits heredity.



Q5 OR. Cytokinesis in plant and animal cells

- In an animal cell, an actin-myosin contractile ring forms at the equator. A cleavage furrow deepens from outside inward until two cells separate (centripetal).
- In a plant cell, the rigid wall prevents furrowing. Golgi vesicles collect at the equator, fuse into a cell plate, and the plate grows from centre outward (centrifugal). Its membranes form two plasma membranes and contents form middle lamella/new wall.
- Both processes partition cytoplasm and organelles after karyokinesis, producing separate daughter cells.



FINAL EXAM CHECKLIST

- Read the command words: identify/name needs a brief direct answer; explain/describe requires linked reasoning; differentiate is clearest in paired points.
- For a 3-mark response, aim for three correct, distinct marking points. For 6 marks, use headings, sequence and a labelled diagram/equation where requested.
- Label diagrams with straight ruled lines; do not let label lines cross. Write biological spellings accurately.
- In Section A, fill only one bubble per question and avoid overwriting. In Sections B and C, attempt the required choice in the exam - this guide includes both choices for preparation.
- Use key terms shown in bold, but write complete sentences where an explanation is asked.

End of complete solutions